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Episode 4 · September 12, 2026 · 3 papers

Fusion Estimation in Multi-sensor Systems for Data Packets with Disrupted Identities

[SKIM](#)

Haoyuan Xu, Yuzhe Li

If your fusion center can't tell which sensor sent which measurement, this paper works out how to estimate anyway by treating the mix-up as an unknown permutation and either averaging over all of them or betting on the most likely one.

PROBLEM Standard multi-sensor fusion assumes every data packet arrives correctly labelled with its source. When identifiers are missing, misallocated, or scrambled by protocol defects, the fusion center silently applies the wrong measurement model to each component and the estimate degrades. The gap is that the shuffle is a discrete combinatorial nuisance variable, not additive noise, so ordinary Kalman-style machinery does not cover it.

RESULTS The abstract offers no numbers at all. It says only that numerical simulations validate the results, with no dataset, no baseline comparison, no error figures, and no statement of how many sensors or how large the permutation group was. The real evidence on offer is theoretical: bounded expectation error for both algorithms and a convergence proof for the expectation maximization step.

METHOD They model the identity scramble as an element of a symmetry group acting on the data vector, then build sufficient statistics so the estimator does not need the full history. Two estimators follow: a Bayesian one that weights each candidate arrangement by its posterior probability, and a greedy one that commits to the single most likely arrangement. Both are proven to have bounded expectation error. When the prior over arrangements is unknown, they learn it with expectation maximization and prove the iteration converges to the right thing.

CAVEAT The permutation group grows factorially in the number of sensors, and the abstract never says how the Bayesian estimator stays tractable past a handful of channels, nor what the greedy estimator's error bound actually is relative to the Bayesian one. Everything is simulation, and the failure mode is motivated by device identity confusion rather than measured against any real protocol trace where identity loss actually occurs. There is also no adversary here: the permutation is assumed random, not chosen by an attacker.

WHY IT MATTERS Peripheral, but with one real hook. Your collective perception work depends on knowing which vehicle produced which detection, and misbehaviour detection assumes the sender label means something. This paper is the clean theoretical statement of what fusion looks like when that label is gone. It is not written for V2X, it does not model channel loss or latency, and the random permutation assumption is weaker than the spoofing case you would actually worry about. Worth knowing it exists as a formal tool if you ever need to argue about identity integrity in a fusion pipeline. Not worth an afternoon otherwise.

arXiv:2609.11062v1 PDF [incremental](#)

Predefined-Time Leaderless Consensus Under Denial-of-Service Attacks

SKIM

Lohitvel Gopikannan, Shashi Ranjan Kumar, Abhinav Sinha

A sliding-mode consensus protocol for second-order agents that keeps a guaranteed convergence deadline even when the communication graph is jammed, by freezing its internal clock during each blackout and restarting it when links come back.

PROBLEM Predefined-time consensus results promise a settling time you fix in advance and that does not depend on initial conditions, but that promise quietly assumes the agents can keep talking. Under denial-of-service jamming the links go dark, the time-scaling function that drives predefined-time convergence blows up as the deadline approaches with no information arriving, and the guarantee breaks.

RESULTS The abstract gives no numbers at all. There is no simulation setup, no agent count, no attack duty cycle, no comparison against an existing resilient consensus scheme. The entire claimed evidence is the analytical statement that consensus is reached by the prescribed duration plus cumulative attack time despite bounded disturbances.

METHOD They build a switching sliding-mode protocol for leaderless second-order nonlinear agents on a strongly connected directed graph. When communication is up, each agent uses relative position and velocity from its neighbours; when links are down, it falls back to local velocity feedback only. The key move is the time-scaling function: it is held constant for the whole duration of a blackout and resumes evolving only once communication is restored, so the clock that drives convergence counts attack-free time rather than wall-clock time. A Lyapunov analysis then gives sufficient gain conditions under standard bounds on attack duration and attack frequency, and the result is a realistic settling time equal to the prescribed duration plus the total attack time accumulated before it.

CAVEAT The settling-time guarantee is real but weaker than the headline suggests, because the bound stretches by the total attack duration, so an adversary who jams for a large fraction of the time pushes the deadline out proportionally and the predefined part of predefined-time becomes a design parameter rather than a delivered deadline. The result also leans on the usual average dwell time style bounds on attack duration and frequency, which an attacker has no obligation to respect, and on every agent detecting the blackout instantaneously so the switch to local velocity feedback is perfectly synchronised. There is no treatment of partial link failure, where some neighbours are reachable and others are not, and nothing in the abstract indicates simulation, let alone hardware.

WHY IT MATTERS Peripheral, with one useful piece. Consensus on an abstract double integrator is not maneuver coordination: platoon formation, merge negotiation and trajectory agreement carry collision constraints, heterogeneous dynamics and explicit intent semantics that position and velocity agreement does not capture, and nothing here touches collective perception or intersection slot allocation. The idea worth stealing is the frozen clock. If you design a cooperative maneuver with a deadline, say a merge that must complete before a gap closes, then holding your convergence schedule constant during a communication outage and resuming it on recovery is a cleaner degradation story than either aborting or letting the deadline expire while blind. Treat it as a technique to borrow for the communication-reliability half of your supporting work, not as a paper about cooperative driving.

CARLaverse: A Highly Modular, Distributed, and Multimodal Framework for Human-in-the-Loop Simulation

SKIM

Patrick Rebling, Philipp Nenninger, Reiner Kriesten

They wired a driving simulator, a bike simulator, and a pedestrian walking rig into one shared CARLA world, and the trick that makes it work is running each participant's own vehicle physics and force feedback locally instead of over the network.

PROBLEM Multi-user human-in-the-loop simulators choke when haptic control loops, which need updates at hundreds of hertz, have to wait on a central server across a network. That kept mixed-traffic studies with real humans playing drivers, cyclists and pedestrians at the same time from being physically convincing.

RESULTS The abstract reports no numbers at all. No latency measurements, no update-rate figures, no user study, no comparison against an existing multi-agent HITL platform. It is an architecture and a code release, and any evidence that the decoupling actually holds up must be somewhere in the nineteen pages.

WHY IT MATTERS Peripheral, with one narrow hook. Nothing here touches maneuver coordination messages, collective perception fusion, or intersection protocols, and the distributed physics is solving a haptic immersion problem rather than a V2X one. The hook is that if you ever want to evaluate a cooperative maneuver negotiation scheme against real human cyclists and pedestrians rather than scripted agents, this is an open-source rig on CARLA that puts a human in each of those roles. Treat it as tooling you might borrow, not a result you need to know.

METHOD The core move is splitting the physics. Latency-critical ego dynamics and high-frequency force feedback run locally on each client node, while a central CARLA server owns the NPC physics and the global traffic state. On top of that sits a modular hardware abstraction layer so the three rigs, called DrivoCARLA, CycloCARLA and WalkoCARLA, plug into the same environment, with the stated goal of cross-institutional experiments.

CAVEAT A reviewer would press hard on the consistency cost of the split. If each client integrates its own ego dynamics locally, the server's view of that vehicle is by definition stale or reconciled, and the paper's own framing does not say how divergence between the local ego state and the shared world is resolved, or what happens to a collision between two locally simulated humans. Claiming cross-institutional scalability without a measured latency budget over a wide-area link is the specific gap.

arXiv:2609.11478v1 PDF incremental